Project	**Mansourah Massarah Gold Project**
Customer:	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title:	**CALCULATION DOCUMENT** **49EE SEWAGE TREATMENT PLANT** **STORM WATER DRAINAGE DESIGN**

Document Number:	**MD-216-49EE-EG-CV-CAL-1058**
Revision:	**D01**
Revision Date:	**26.12.2023**
Status:	**AS BUILT**

Outotec Document Number:	900707	GAU01	AG90	CED01	50004	OU100706311	3
L&T Document Number:	O19045	910	C	K	24362	001	3

REVISION HISTORY

Rev	Date	Revision Text	Prepared By	Reviewed By	Approved By	Employer
A01	01.12.2020	FIRST ISSUE	APEX	BJB	SJM	
C01	09.01.2021	ISSUED FOR CONSTRUCTION	APEX	BJB	SJM	
C02	18.01.2021	ISSUED FOR CONSTRUCTION & REVISED AS MARKED	APEX	BJB	SJM	
D01	26.12.2023	AS BUILT	APEX	BJB	SJM	

TABLE OF CONTENTS

DEFINITIONS AND ABBREVIATIONS.....3

1. INTENT.....4
2. EXECUTIVE SUMMARY.....4
3. SCOPE.....5
4. STORMWATER SYSTEM.....6
   - 4.1. Main Design Criteria.....6
   - 4.2. Calculation Results.....10

References:.....14

ATTACHMENT 1: SKETCH FOR CATCHMENT AREA.....15

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

DEFINITIONS AND ABBREVIATIONS

Document	:	Information and the medium on which it is contained (ISO 9000:2015)
EMPLOYER	:	'EMPLOYER' means MA'ADEN GOLD AND BASE METALS (MGBM) and the PMC American Bechtel Inc. or their representatives
Contractor	:	Contractor' means consortium composed of LARSEN & TOUBRO LIMITED, Saudi Arabia AND OUTOTEC SAUDI ARABIA LLC, Saudi Arabia (for IK Contract) and LARSEN & TOUBRO LIMITED – ECC Construction Group – (SHJ BR) and OUTOTEC ENGINEERING DMCC, Dubai (for OOK Contract).
'PMT'	:	Ma'aden Project Management Team
'MMGP Project'	:	Mansourah Massarah Gold Processing Project
Attribute	:	Information element that describes a document or tag.
Supplier	:	An entity (supplier, consultant, sub-contractor) under the direct project management of Contractor
Sub-Supplier	:	A sub entity (supplier, consultant, sub-contractor) under the direct project management of Contractor.
QA	:	Quality Assurance
QC	:	Quality Control
Works	:	Full scope of the EPC Contract including the engineering, procurement, construction, commissioning and associated activities

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

1. INTENT

This document will list the main design criteria and design calculations relevant to the following components of the Sewage treatment plant area of Mansourah & Massarah Gold project (MMGP):

- The Sewage treatment plant stormwater system composed of open channels and pipes
- The stormwater pipes discharging some of the watersheds to the outside of the project into the natural outfalls.

2. EXECUTIVE SUMMARY

The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah & Massarah (MMGP) Gold Project.

MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah-Massarah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah - Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim.

The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week.

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

3. SCOPE

The design document has been prepared for Sewage Treatment Plant Drainage System (Drawing number MD-216-49EE-EG-CV-GAD-1585) based on Overall Project Layout drawing MD-216-1000-EG-ME-PPD-1001 relevant ground terracing has been obtained from site leveling drawing and necessary rainfall intensity has been captured from relevant design document.

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

4. STORMWATER SYSTEM:

The stormwater system is formed of open trapezoidal channels discharging into the natural ground. These channels are connected via concrete pipes at locations that are crossing the roads or where vehicular access is expected.

4.1 Main Design Criteria

Rainfall Analysis:

The rainfall intensities and rainfall depths values were taken from the generated Rainfall Depths Curves that was indicated in approved document no. MD-216-4400-EG-ME-CAL-1001. This Rainfall Depths curve is shown in Figure 1 below:

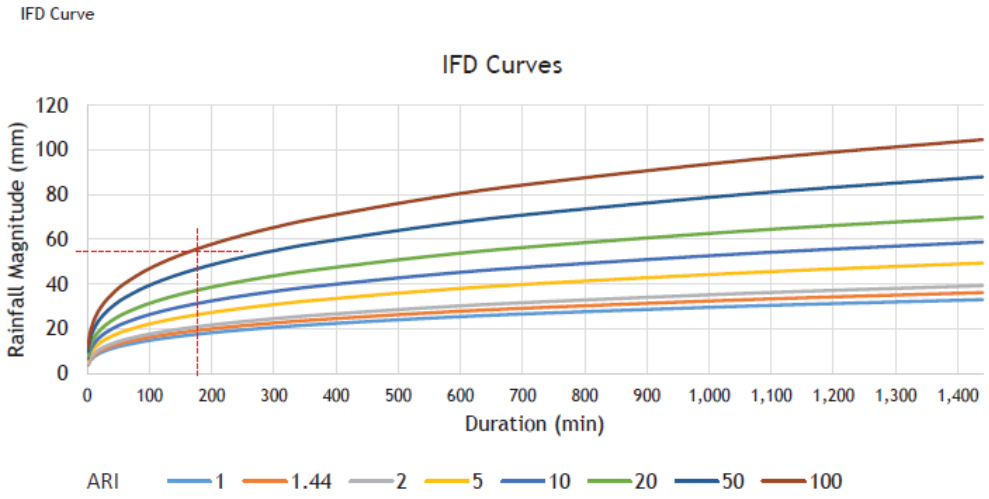


Figure 1 Rainfall Depths Curves

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Table 1 below lists the derived rainfall intensity values for the 10-year return period that will be adopted in the design.

Table 1 Rainfall Intensity Values

10-year storm	10 min	20 min	30 min	60 min	120 min	180 min	360 min	720 min	1440 min
I (mm/hr)	79.58	48.9	36.8	22.7	14.0	10.5	6.5	4.0	2.5

Time of Concentration:

The time of concentration of each watershed is calculated using Kirpich Formula as follows:

$$T_c = 0.0195\left(\frac{L}{S^{0.5}}\right)^{0.77}$$

Whereas:

T_C = Time of concentration in minutes

L = Length of stream, m

S = Average slope of the stream m/m

Since most of these watersheds are relatively small, all time of concentrations are below 10 minutes, so a minimum value of 10 is adopted.

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Peak Flow Calculations:

The peak flow is calculated for each watershed using Rational Method.

The rational method is recommended for catchment areas less than or equal to 81 hectares which is the case in our project. It is an empirical formula relating rainfall intensity to runoff, and yields a peak discharge. It is usually used to calculate the peak discharge from various drainage basins as follows:

$$Q = \frac{C.I.A}{360}$$

Where:

- Q is the peak discharge, m^3/s;
- I is the average rainfall intensity, mm/hr, for a duration equal to the time of concentration and for a selected design storm;
- A is the drainage area, ha.
- C is the runoff coefficient representing land use characteristics of the watershed. Since this is an industrial area, a runoff coefficient of 0.75 is assumed.

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Hydraulic Design:

The hydraulic design of the network is done using Manning's equation, as follows:

$$V = \frac{1}{n} R^{2/3} S^{0.5}$$

Where

V, is the mean velocity of flow, in m/s;

n, is the Manning's roughness coefficient for open channel flow, n is assumed to be 0.025 for earth open channels and 0.013 for concrete pipes

R, is the hydraulic radius in m; and

S, is the slope of energy grade line, or channel bottom slope, in m/m.

The capacity of the open channel is determined from the continuity equation:

$$Q = A.V$$

Where Q is the flow rate in m^3/s,

V is the velocity in m/s, and

A is the flow area in m^2.

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

C02

4.2 Calculation Results

The following tables list the hydraulic results for the different components of the stormwater network.

Table 2 below lists the results of the watershed flows and characteristics.

Table 2 Watershed Results

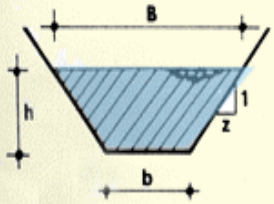
Area (A) mkd.	Drain mkd.	Length of stream (m)	Time of concentration Tc (min)	Area (ha)	Runoff coefficient `C'	Rainfall Intensity `I' (mm/hr)	Runoff from catchment (Litre/sec)	Direct input		Total Flow (Litre/sec)
								From drain mkd.	Quantity (Litre/sec)	
STP1	D1	37	10.00	0.0365	0.75	79.58	6.05			6.05
STP2	D2	9.3	10.00	0	0.75	79.58	0.00	D1	6.05	6.05
STP3	D3	13	10.00	0.0145	0.75	79.58	2.40			2.40
STP4	D4	8.865	10.00	0.0204	0.75	79.58	3.38			3.38
STP5	D5	9.3	10.00	0	0.75	79.58	0.00	D3 & D4	5.79	5.79
STP6	D6	12.6	10.00	0.0294	0.75	79.58	4.87			4.87
STP7	D7	18.84	10.00	0.0218	0.75	79.58	3.61	D5 & D6	10.66	14.27
STP8	D8	22.5	10.00	0.0394	0.75	79.58	6.53	D2 & D7	20.33	26.86
STP9	D9	9.55	10.00	0	0.75	79.58	0.00	D8	26.86	26.86
STP10	D10	14.52	10.00	0.0378	0.75	79.58	6.27			6.27
STP10a	D10a	13.5	10.00	0	0.75	79.58	0.00	D10	6.27	6.27
STP11	D11	8.1	10.00	0.0274	0.75	79.58	4.54			4.54
STP12	D12	9.38	10.00	0	0.75	79.58	0.00	D10a & D11	10.81	10.81
STP13	D13	7.8	10.00	0.0041	0.75	79.58	0.68			0.68
STP14	D14	52.11	10.00	0.025	0.75	79.58	4.14	D13	0.68	4.82
STP15	D15	48.3	10.00	0.0271	0.75	79.58	4.49	D14	4.82	9.32
STP16	D16	27.8	10.00	0.0614	0.75	79.58	10.18	D15 & D22	36.57	46.75
STP17	D17	42	10.00	0.0455	0.75	79.58	7.54			7.54
STP18	D18	17	10.00	0	0.75	79.58	0.00	D17	7.54	7.54
STP19	D19	37.18	10.00	0.0205	0.75	79.58	3.40	D18	7.54	10.94
STP20	D20	14.93	10.00	0.0055	0.75	79.58	0.91	D19 & D21	22.22	23.13
STP21	D21	7.08	10.00	0.0028	0.75	79.58	0.46	D12	10.81	11.27
STP22	D22	6.12	10.00	0.0024	0.75	79.58	0.40	D9	26.86	27.26

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Channel sizing

Channel Section	Wetted area	Wetted perimeter	Hydraulic radius (R)	Water top width (m)
	$(b + zh)\,h$	$b + 2h\sqrt{1+z^2}$	$\frac{(b + zh)\,h}{b + 2h\sqrt{1+z^2}}$	$b + 2zh$

Where:

b = Bottom width in m..

a = Top water width in m.

h = Depth of water in m.

z = Side slope

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Table 3 Channels Hydraulic Results

Area (A) mkd.	Drain mkd.	Size of Channel							Slope	Velocity (m/sec.)	Capacity (Qcap.) (lit./sec.)
		Bottom width 'b` (m)	Depth (m)		Side slope		Top width (m)				
			Final depth (m)	Water depth 'h` (m)	H (z)	V	Top water width 'a` (m)	Top width (m)			
STP1	D1	0.3	0.4	0.35	1	1	1	1.1	0.0033	0.723	164.40
STP2	D2	Road crossing drainage pipe									
STP3	D3	0.3	0.4	0.35	1	1	1	1.1	0.004	0.796	181.00
STP4	D4	0.3	0.395	0.345	1	1	0.99	1.09	0.005	0.883	196.48
STP5	D5	Road crossing drainage pipe									
STP6	D6	0.3	0.425	0.375	1	1	1.05	1.15	0.0067	1.067	270.08
STP7	D7	0.3	0.45	0.4	1	1	1.1	1.2	0.00167	0.551	154.23
STP8	D8	0.3	0.5	0.45	1	1	1.2	1.3	0.002	0.641	216.40
STP9	D9	Road crossing drainage pipe									
STP10	D10	0.3	0.345	0.295	1	1	0.89	0.99	0.0033	0.662	116.25
STP10a	D10a	Road crossing drainage pipe									
STP11	D11	Road crossing drainage pipe									
STP12	D12	Road crossing drainage pipe									
STP13	D13	0.3	0.3	0.25	1	1	0.8	0.9	0.0025	0.530	72.92
STP14	D14	0.3	0.45	0.4	1	1	1.1	1.2	0.0025	0.674	188.71
STP15	D15	0.3	0.55	0.5	1	1	1.3	1.4	0.0025	0.758	303.22
STP16	D16	0.3	0.6	0.55	1	1	1.4	1.5	0.00167	0.652	304.83
STP17	D17	0.3	0.375	0.325	1	1	0.95	1.05	0.002	0.542	110.02
STP18	D18	Road crossing drainage pipe									
STP19	D19	0.3	0.475	0.425	1	1	1.15	1.25	0.002	0.622	191.72
STP20	D20	0.3	0.6	0.55	1	1	1.4	1.5	0.008	1.427	667.18
STP21	D21	0.3	0.475	0.425	1	1	1.15	1.25	0.0057	1.050	323.65
STP22	D22	0.3	0.55	0.5	1	1	1.3	1.4	0.0033	0.871	348.37

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Table 4 below lists the results of the concrete pipes hydraulic analysis.

Table 4 Concrete Pipes Hydraulic Results

Area mkd.	Pipe Drain mkd.	Dia. of pipe`D' (m)	Slope	Velocity (m/sec.)	Capacity Qcap. (lit./sec.)	Nos. of Pipe	Qcap. (lit./sec.)
STP2	D2	0.3	0.005	1.10	62.38	1	62.38
STP5	D5	0.3	0.0033	0.89	50.68	1	50.68
STP9	D9	0.3	0.004	0.98	55.80	1	55.80
STP10a	D10a	0.3	0.0033	0.89	50.68	1	50.68
STP11	D11	0.2	0.01	1.18	29.92	1	29.92
STP12	D12	0.3	0.0057	1.17	66.61	1	66.61
STP18	D18	0.3	0.002	0.69	39.45	1	39.45

Considering pipe flowing at 75% full.

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

REFERENCES

Reference is made to codes, standards and specifications as indicated in Civil, Structural and Architectural Design Criteria (Doc. No. MD-216-1000-EG-ST-CRT-1001) Clause 4.1

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

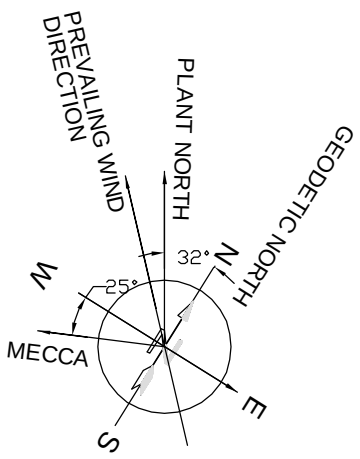
C02

ATTACHMENT 1: SKETCH FOR CATCHMENT AREA

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.



SKETCH FOR CATCHMENT AREA

